Final Exam

● Graded

Student

Xiaohan Guo

Total Points

43.5 / 45 pts

Question 1

Overall Grades **43.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 1.5 pts** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Xiaohan Guo	xg113	郭潇涵

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: 郭潇涵

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	45	5	5	4	5	5	5	5	43.5

Problem 1. [5'] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$

write in matrix form:

$$\begin{bmatrix} 2 & -3 & 0 \\ 0 & -2 & -3 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow$$

Augmented matrix

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \quad R_3 \leftarrow R_3 - \frac{1}{2}R_1$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 3/2 & 2 & 1/2 \end{array}\right] \quad R_3 \leftarrow R_3 + \frac{3}{4}R_2$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 0 & -1/4 & 5/4 \end{array}\right]$$

$$-\frac{1}{4}z = \frac{5}{4} \qquad z = -5$$

$$-2y - 3z = 1 \qquad -2y = 1 + 3z \qquad y = -\frac{1+3z}{2} = -\frac{1-15}{2} = 7$$

$$2x - 3y = 1 \qquad 2x = 1 + 3y \qquad x = \frac{1+3y}{2} = \frac{1+21}{2} = 11$$

Solution:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$$

4.5/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

$A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_4 \leftrightarrow R_1} \begin{bmatrix} 1 & 3 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 2 & 4 & 2 \end{bmatrix} \xrightarrow{\substack{R_2 \leftarrow R_2 + 5R_1 \\ R_4 \leftarrow R_4 - 2R_1}} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 15 & 15 \\ 0 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix} \xrightarrow{\substack{R_2 \leftarrow \frac{R_2}{15} \\ R_4 \leftarrow \frac{1}{2}R_4}}$

$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - 3R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ RREF

(1) pivot column $= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

pivot basis of column space A: $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

(2) pivot basis of row space of A: $\left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}^T, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}^T \right\}$ -0.5

(3) Null space $Ax = 0$

$\begin{cases} x - z = 0 \\ y + z = 0 \end{cases}$ let $z = t$

$y + z = 0 \quad y = -z = -t$

$x - z = 0 \quad x = z = t$

General solution: $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} t$

Basis of the null space of A: $\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

Problem 3. [5] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$\because T: \mathbb{R}^3 \to \mathbb{R}^2 \quad \therefore A$ should be a 2×3 matrix

$$A\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$$

$$A\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix} \Rightarrow A\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$$

$$A\begin{bmatrix}0\\0\\-1\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$$

let $\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = B$.

$R_2 \leftarrow -R_2$
$R_1 \leftrightarrow R_3$

$$R = \begin{bmatrix}1&1&1\\1&0&0\\0&1&0\end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc}1&1&1&1&0&0\\1&0&0&0&1&0\\0&1&0&0&0&1\end{array}\right] \quad R_2 \leftrightarrow R_3 \quad \left[\begin{array}{ccc|ccc}1&1&1&1&0&0\\0&1&0&0&0&1\\1&0&0&0&1&0\end{array}\right]$$

$R_3 \leftarrow R_3 - R_1$
$R_1 \leftarrow R_1 - R_2$

$$\left[\begin{array}{ccc|ccc}1&0&1&1&0&-1\\0&1&0&0&0&1\\0&1&-1&-1&1&0\end{array}\right] \quad R_3 \leftarrow -R_3 + R_2 \quad \left[\begin{array}{ccc|ccc}1&0&1&1&0&-1\\0&1&0&0&0&1\\0&0&1&1&-1&-1\end{array}\right] \quad R_1 \leftarrow R_1 - R_3 \quad \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&0&0&1\\0&0&1&1&-1&-1\end{array}\right]$$

$$B^{-1} = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$$

$$AB = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$$

$$ABB^{-1} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}B^{-1}$$

$$A = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}$$

$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$ $P_1 \leftarrow P_1 - P_2$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$ det不变

Problem 4. [5] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$ Suppose λ is eigenvector of A.

$\det(A-\lambda I) = 0$ $A - \lambda I = \begin{bmatrix} 1-\lambda & 0 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$

$\det = (1-\lambda)^2(2-\lambda) = 0$ $\lambda_1 = 1$, $\lambda_2 = 2$

② For $\lambda_1 = 1$ $AM = 2$ $GM = 1$

$A - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix}$

~~let $\vec{v}$ be eigenvector of $\lambda = 1$~~

$(A-I)\vec{v} = 0 \Rightarrow y = 0$

~~There does not exist a eigenvector in $\lambda = 1$.~~

~~let $\vec{v_1} = [0, 0, t]^T$ be eigenvector in $\lambda = 1$~~

① For $\lambda_2 = 2$ $AM = 1$ $GM = 1$ $AM = GM$.

$A - 2I = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix}$

let $\vec{v_1}$ be eigenvector of $\lambda_2 = 2$

$(A-2I)\vec{v_1} = 0$

$-x = 0$

$-2x - 2y - z = 0$ $\Rightarrow$ $x = 0$, let $z = t$, $-2y = z$, $y = -\frac{1}{2}z = \frac{t}{-2}$

$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{t}{-2} \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -\frac{1}{2} \end{bmatrix} t$ $\vec{v_1} = \begin{bmatrix} 0 \\ 1 \\ -\frac{1}{2} \end{bmatrix}$

let $\vec{v_2}$ be eigenvector, Jordan form

$(A-I)\vec{v_2} = \vec{v_1}$ $(x, y, z) = (-3/4, 1, t)$ $\vec{v_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ -2 & -2 & 0 & -\frac{1}{2} \end{array}\right]$ $y = 1$, $-2x - 2y = -\frac{1}{2}$, $z = t$ #

$-2x = -\frac{1}{2} + 2y = \frac{3}{2}$, $x = -3/4$

$\vec{v_2} = \begin{bmatrix} -3/4 \\ 1 \\ 0 \end{bmatrix}$

$J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$P = \begin{bmatrix} 0 & -3/4 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -1/2 \end{bmatrix}$

4/5

Problem 6. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$$f(x,y,z) = A\bar{v} \qquad A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

$$\det(A-\lambda I) = \begin{vmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{vmatrix} = (3-\lambda)\begin{vmatrix} 3-\lambda & -1 \\ -1 & 4-\lambda \end{vmatrix} + \begin{vmatrix} 0 & 3-\lambda \\ 1 & -1 \end{vmatrix}$$

$$= (3-\lambda)\cdot\left[(3-\lambda)(4-\lambda)-1\right] + 0 - (3-\lambda)$$

$$= (3-\lambda)(11-7\lambda+\lambda^2) - (3-\lambda)$$

$$= (3-\lambda)(10-7\lambda+\lambda^2)$$

$$= (3-\lambda)(\lambda-2)(\lambda-5)$$

$\lambda_1 = 3$, $\lambda_2 = 2$, $\lambda_3 = 5$.

for $\lambda_1 = 3$, let v_1 be eigenvector $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} v_1 = 0$ $\begin{matrix} z = 0 \\ -z = 0 \\ x-y+z = 0 \end{matrix}$ $\Rightarrow$ $\begin{matrix} x = t \\ y = t \\ z = 0 \end{matrix}$

$v_1 = (1,1,0)$

for $\lambda_2 = 2$, let v_2 be eigenvector $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} v_2 = 0$ $\begin{matrix} x+z = 0 \\ y-z = 0 \\ x-y+2z = 0 \end{matrix}$ $\Rightarrow$ $\begin{matrix} x = t \\ y = -t \\ z = -t \end{matrix}$

$v_2 = (1,-1,-1)$

for $\lambda_3 = 5$, let v_3 be eigenvector $\begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} v_3 = 0$ $\begin{matrix} -2x+z = 0 \\ -2y-z = 0 \\ x-y-z = 0 \end{matrix}$ $\Rightarrow$ $\begin{matrix} x = t \\ y = -t \\ z = 2t \end{matrix}$

$v_3 = (1,-1,2)$

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 0 & -1 & 2 \end{bmatrix} \qquad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$A = PDP^{-1}$

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

Since $W = aV_1 + bV_2 + cV_3$

write in matrix form

$$\begin{bmatrix} 0 & 2 & -1 \\ 2 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \begin{array}{l} R_2 \leftarrow R_2 + R_2 \\ \curvearrowright \\ R_2 \leftrightarrow R_1 \end{array} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right] \begin{array}{l} R_3 \leftarrow R_3 - \frac{1}{2}R_2 \\ \curvearrowright \end{array} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & 3/2 & 3/2 \end{array}\right]$$

$$\begin{cases} 2a - b + c = 8 \\ 2b - c = -3 \\ 3/2\, c = 3/2 \end{cases} \Rightarrow \begin{array}{l} 2a = 8 + b - c = 8 - 1 - 1 = 6 \quad a = 3 \\ 2b = -3 + c = -2 \quad b = -1 \\ c = 1 \end{array}$$

Solution: $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

span the whole subspace

~~V = S~~ for subspace V, $\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$ They are LI and can should be its basis

① get orthognal basis, use Gram-schmidt rule

$v_1 = [0\ 1\ 1\ 0]^T$, $v_2 = [3\ 0\ 2\ -1]^T$.

suppose ~~u~~ $v_1 = u_1$

$$u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} \cdot u_1$$

$$= \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2} \cdot \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$$

now, we get u_1, u_2 they are orthogonal ~~and~~ basis

② Orthonormal

$\|\vec{u_1}\| = \sqrt{0^2+1^2+1^2+0^2} = \sqrt{2}$ $\quad \vec{u_1}' = \frac{1}{\sqrt{2}}\vec{u_1} = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)$

$\|\vec{u_2}\| = \sqrt{3^2+1^2+1^2+1^2} = \sqrt{12} = 2\sqrt{3}$ $\quad \vec{u_2}' = \frac{1}{2\sqrt{3}}\vec{u_2} \neq$ ~~1/√3~~

$= (\frac{3}{2\sqrt{3}}, -\frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}}, -\frac{1}{2\sqrt{3}})$

orthonormal Basis: $\left(\begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}, \begin{bmatrix}\frac{3}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\end{bmatrix}\right)$.

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

matrix A.

for square $A^T = -A$, for $n \in \mathbb{R}$?

$a_{nn} = -a_{nn}$ $\quad a_{nn} = 0$

the entry on the diagonal of A are all 0

$\det(A^T) = \det(A)$, ~~$A^T = -A$~~,

$A^T = -A \Rightarrow \det(-A) = \det(A)$

$\det(-A) = \det((-1)A) = (-1)^n \det(A)$

$= \det(A)$

if n is odd, $(-1)^n = -1$

$-\det(A) = \det(A) \Rightarrow \det(A) = 0$.

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

LHS $(\|\vec{u}+\vec{v}\|)^2 = \|\vec{u}\|^2 + 2\vec{u}\vec{v} + \|\vec{v}\|^2$

RHS $(\|\vec{u}-\vec{v}\|)^2 = \|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2$

RHS = LHS , $\|\vec{u}\|^2 + 2\vec{u}\vec{v} + \|\vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u}\vec{v} + \|\vec{v}\|^2$

$2\vec{u}\vec{v} = -2\vec{u}\vec{v}$

$\vec{u}\cdot\vec{v} = 0$

$\theta = \langle \vec{u}, \vec{v} \rangle$

$\cos\theta = \dfrac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|\cdot\|\vec{v}\|} = 0 \Rightarrow \theta = 90^\circ$

$\therefore \vec{u} \perp \vec{v}$.